

EVSP–DR: where the research stands

Updated 16 September 2026, 21:25 EDT; earlier experiment tables retain their stated snapshot.

New proof: [Time-only fleet bounds for all 102 instances](#) · [Full-Partille job and memory rationale](#).

In one paragraph

We optimize electric-bus schedules and charging for Transdev's Partille network (GIRO data) with column generation (CG) and a MIP, under time-varying electricity prices. **Fleet size:** in 93 of 102 test instances, our lower bound equals GIRO's own bus count, so GIRO's fleet is already optimal in our model there; we have matched that bound with a valid schedule in 67 of 102 and the rest are search gaps, not model gaps. In 9 instances the bound is one bus *below* GIRO — the only place our method could beat GIRO on fleet — and none has been closed yet.

Charging cost: on one 5-duty instance, re-optimizing charging alone cuts GIRO's electricity bill by ~44%; also re-optimizing which bus does which trip saves a further 2–7%. Both numbers were obtained without GIRO's 15% reserve and 3-minute charging rules, and a rerun with those rules is in progress. **Next 48 hours** decide three things: whether the 1-bus savings are real, whether the charging gain survives realistic rules, and whether the warm-start method matters or the MIP just needed more time.

Question 1 — Can we recover GIRO's fleet, and can we beat it?

Answer so far: GIRO is fleet-optimal in our model in 93 of 102 instances; we match the lower bound in 67; possible 1-bus savings exist in 9 instances (chains 4 and 5, $k \geq 27$) and are unproven.

	Count
Instances (6 chains $\times$ $k = 16 \dots 32$)	102
Lower bound = GIRO's k	93
Lower bound = $k - 1$ (model could beat GIRO)	9
Bound matched by a validated schedule (including 3 h searches)	67
GIRO fleet count matched (a different test)	70
Open gap to the lower bound	35
Open: bound k , best schedule $k+1$ or worse	26
Open: bound $k-1$, best schedule k or worse	9

Every one of the 128 schedules passes physical replay. Duplicate trip copies are driven empty; this leaves fleet and cost unchanged. The bounds are numerical certificates on the discretized model (2.5 kWh / 5 min), not exact-arithmetic proofs.

Important context: all 42 GIRO duties are feasible in our model once their charging is re-planned (0 of 42 are feasible with GIRO's own charging plan, under the baseline 240 kW replay). This establishes a continuous-charging k-bus upper bound. It does not itself establish representability on our event grid; that requires a separate feasible event-grid schedule.

Vehicle-group result: time-only scheduling with deadheading needs exactly GIRO's fleet in all 102 instances when groups stay separate, for each group individually. With mixing allowed, the time-only minimum is k in 93 cases and k-1 in the same nine as our LP results. Thus group separation rules out those one-bus savings, independently of energy constraints. The electric-bus integer saving with mixing is still open. [All 102 saved LP endpoints](#).

Open: can an integer electric-bus schedule achieve the one-bus saving when vehicle groups may mix? The six single-factor arms also test the effects of depot power, reserve and battery size.

Question 2 — Does warm-starting from the previous instance matter?

Answer so far: unresolved. The earlier "fresh 6/24 vs warm 24/24" result mostly measured MIP time, not CG.

Fresh and warm CG reach the identical LP bound in all 24 comparisons. Fresh MIPs then missed the target under a 30-minute fleet search; only 4 of 18 misses are proven pool limitations, the rest are timeouts. Whether a longer search closes them is running now. Also running: a chain built from *random* trip groups rather than GIRO's duties, to test whether warm-starting helps without knowing GIRO's grouping.

Question 3 — What is demand response worth?

Answer so far: on one 5-duty instance, most of the value is smarter charging; re-routing adds 2–7% on top. Not yet shown under realistic charging rules or at larger size.

Tariff peak	GIRO as-is	Same duties, optimized charging	Optimized duties + charging
08:00	230.3–231.0	128.3	124.7
12:00	289.6–290.6	164.2	160.6
18:00	223.4–223.7	95.3	88.3

Five buses in every arm; ending energy matched between the two optimized arms. **Caveats that matter:** both optimized arms run batteries to ~0% (GIRO never goes below 19%) and use charges as short as 26 seconds (GIRO's minimum is 3 min). Part of the 44% is spending GIRO's safety margin. Settings: 240 kWh / 350 kW / no start fee / flat depot rate.

Running: the same three arms at $k=5$ with a 15% reserve and 3-minute minimum (decisive for whether the gain is real), and fresh CG vs fixed-duty at $k=15$ under three synthetic tariffs plus one real SE3 price day.

Question 4 — Which algorithm changes helped?

Event-based time/SOC graphs with exact pricing (summer) give certified LP bounds where the old capped search could not. Indexed route replay: -12 to -18% CG time. Skipping unused LP-matrix construction: -9 to -15%. Inheriting the full column pool instead of 512 routes: -41 to -65% CG time and better fleets (9→8, 11→10, 17→15) at the same LP value. Three inputs, two orders each — directional, not statistical. Capacity-pricing caching: no measurable effect (both arms time out).

Our model vs GIRO's reality

	Our baseline	GIRO / Transdev	Status
Battery	240 kWh, one type	236.4 / 239.0 kWh, two vehicle groups	Negligible
Terminal charging	240 kW constant	371→120 kW, SOC-dependent	Ours is slower on average
Depot charging	240 kW	60 kW	Ours is 4× faster — testing (item 11)
Minimum SOC	none	15% hard	Testing (k5 rerun, item 11)
Min charge duration	none	3 min + 45 s setup	Testing (k5 rerun, item 10)
Vehicle groups	mixed	route-21 blocks separate	Testing (item 11)
Shared charger capacity	unlimited	1–2 per station, FIFO at 4808	Not modeled
Tariff in scaling runs	flat	—	$k=15$ tariff runs in progress

Running now

Times are EDT. For running jobs, Expected gives the Slurm allocation deadline (start + wall limit). Sequential projections use the remaining dependency chain at full wall limits, with zero queue delay; they are not scheduled finish times.

Experiment	Answers	Jobs	Expected
Fresh $k=15$ pools, 3 h fleet search × 3 seeds	Q2: is the 6/24 an artifact of MIP time?	18	16 Sep, 22:22

Chain 5 k=31, 12 h search on existing pool	Q1: can we find 30 buses where GIRO has 31?	1	17 Sep, 07:22
k=32 repeats × 3 seeds, chains 1/3/4/5	Q1: how much is seed luck?	12	16 Sep, 22:22 (8 new runs; 4 seed-zero results reused)
k=5 three-arm with 15% reserve + 3-min	Q3: does the DR gain survive real rules?	6	16 Sep, 21:55 (5/6 completed)
k=15 × (3 synthetic + 1 real tariff), CG vs fixed-duty	Q3: does the gain hold at scale?	48	17 Sep, 03:04 fixed / 07:04 fresh CG
Chain 5 strict physics (60 kW depot, 15%, groups) to k=31	Q1: are the 9 cases an artifact?	62 sequential	21 Sep, 13:24 — full-budget dependency projection
Random-trip-group chain to 364 trips	Q2: does warm start need GIRO's grouping?	42 sequential	17 Sep, 22:37 — full-budget dependency projection
Full Partille, 40 duties / 948 trips, 12 h allocation	Scale	CG 343119: Scaglione, 120 GiB; 11 h 45 min CG; checkpoint fallback tested	Waiting for graph 341404_0; no scheduled start
Frölunda ladder	Generalization	1 (48 h)	18 Sep, 18:12
C5 k31 single-factor arms, full-pool initialization	Q1: isolate depot power, reserve, battery size and vehicle groups	6 arms / 750 replay tasks	Replay started 16 Sep, 20:22; full-pipeline finish not scheduled

What is still needed before writing up

Must have

1. Q1: group separation now rules out k−1 in all nine cases. Still open: a validated electric-bus schedule attaining k−1 when groups may mix, and sensitivity to the tighter physics. (Items 8, 11.)
2. Q3 under GIRO's rules: the k=5 three-arm with 15% reserve and 3-min minimum. If the gain collapses, the DR story changes. (Running.)
3. Q3 at scale: at least k=15, at *equal fleet*, on ≥ 3 chains. Needs the k=15 tariff CGs plus a follow-up MIP with fleet cap = k. (Running + planned.)
4. Q2 resolved one way or the other: fresh k15 with 3 h search. (Running.)
5. One large instance: full Partille certified or bounded. (CG queued after graph preparation.)

Should have

6. Shared charger capacity at ≥ 1 station in the $k=15$ runs — the one GIRO constraint we have never modeled.
7. Frölunda beyond $k=2$.
8. A real price series in the headline (currently internal-only under license; need a publishable alternative such as Nord Pool day-ahead if terms allow).

Decisions pending

- Cluster: six full single-factor arms are now launched. Other new submissions remain on hold until the fresh- $k15$, C5 $k31$, constrained- $k5$ and $k32$ seed results are complete.
- Full-Partille CG 343119 is queued on Scaglione with 120 GiB and a 12 h allocation (11 h 45 min CG plus shutdown allowance), after the existing graph job. Memory sizing uses the fresh single-process run, not the older eight-worker inheritance peaks. The downstream MIP remains held.

Sources

Chain table with bounds:

[outputs/independent_review_20260916/execution/audited_chain_results.cs](#)

v. Verdicts and evidence: [execution/README.md](#). Independent review: [REVIEW.md](#).

Charging three-arm: [execution/f6/](#). GIRO replay: [execution/f4/](#). All hashes and job IDs in [execution/ledger.json](#).

[F1–F9 verdicts and execution ledger](#) · [Chain results with numerical lower bounds](#) · [Independent review](#)

[Figures with explanations](#) · [CG curves and bus schedules](#) · [Historical research log](#)